A HAUSDORFF SPACE IS LOCALLY QUASI-COMPACT EXACTLY WHEN EVERY POINT HAS A QUASI-COMPACT NEIGHBOURHOOD

THE STACKS PROJECT AUTHORS

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Allowed standard prerequisites (editorial).

- The statement that a quasi-compact subset of a Hausdorff space is closed, and its refinement form producing disjoint opens around a point and a closed subset missing it. Invoked inside the counted proof (source lines 2247-2248 and 2252-2253). Produces a closed quasi-compact neighbourhood inside the given one.
- The statement that a closed subset of a quasi-compact set is quasi-compact. Invoked at the end of the counted proof (source lines 2255-2256). Makes the closed neighbourhood quasi-compact.

Source reference. Original lemma lemma-closed-in-quasi-compact: sources/stacks-topology/pinned-original-source.tex, source lines 1879–1879. Navigation pointer only: the counted text refers to the Stacks item lemma-closed-in-quasi-compact, which is an existing labeled result of this same chapter and is not part of this card. The pointer carries the target's own label as its number and its definition line as its span; its content is not reproduced here.

Source reference. Original lemma lemma-quasi-compact-in-Hausdorff: sources/stacks-topology/pinned-original-source.tex, source lines 1894–1894. Navigation pointer only: the counted text refers to the Stacks item lemma-quasi-compact-in-Hausdorff, which is an existing labeled result of this same chapter and is not part of this card. The pointer carries the target's own label as its number and its definition line as its span; its content is not reproduced here.

Lemma 0.1. *A Hausdorff space is locally quasi-compact if and only if every point has a quasi-compact neighbourhood.*

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Proof. Let X be a Hausdorff space. Let $x \in X$ and let $x \in E \subset X$ be a quasi-compact neighbourhood. Then E is closed by Lemma lemma-quasi-compact-in-Hausdorff. Suppose that $x \in U \subset X$ is an open neighbourhood of x. Then $Z = E \backslash U$ is a closed subset of E not containing x. Hence we can find a pair of disjoint open subsets $W, V \subset E$ of E such that $x \in V$ and $Z \subset W$, see Lemma lemma-quasi-compact-in-Hausdorff. It follows that $\overline{V} \subset E$ is a closed neighbourhood of x contained in $E \cap U$. Also $\overline{V}$ is quasi-compact as a closed subset of E (Lemma lemma-closed-in-quasi-compact). In this way we obtain a fundamental system of quasi-compact neighbourhoods of x. $\square$